Allowance edges or transfer edges? Evaluating on-chain smart contract reputation out of window against degree baselines

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Abstract

Before using a decentralized finance contract, an owner signs an ERC-20 approval that lets the contract move its tokens. Graph reputation scores could inform that decision, but an existing method, Adaptive Weighted PageRank (AWP), walks transfers, which record payments, not authorizations. We ask what allowance edges add. On public Arbitrum One logs from October 2023 to June 2026 we compared EndorseRank, PageRank on the latest allowances, with AWP and with the raw degrees that both walks smooth, for 15,107 contracts approved by at least three owners on five tokens. Amounts were valued in US dollars after a pre-specified analysis in raw token units was complete; we report this revised analysis, applying the pre-specified rules as a re-analysis. Each score agreed most with checks built from its own edges (Kendall τ 0.585 [95% CI 0.578, 0.592] for EndorseRank, 0.572 [0.564, 0.580] for AWP). With scores frozen before two three-month windows, AWP ranked later new transfer senders above the transfer in-degree ($\Delta\tau$ +0.043 [0.038, 0.048] and +0.022 [0.017, 0.027]), whereas EndorseRank ranked later new approval pairs below the count of approving owners (-0.019 [-0.026, -0.011] and -0.034 [-0.038, -0.029]). These degree comparisons were chosen after the labels were known; a variant with activity restarts that beat the count is a hypothesis. For spenders whose owners are not themselves approved, EndorseRank is a reweighted approval count. On the later liquidation-free close rate of 70 contracts trading on GMX V2, an outcome neither graph contains, no lead of the approval count over the transfer count was resolved ($\Delta\tau$ +0.084 [-0.013, 0.176]; 97.5% interval [-0.026, 0.188]), nor of the activity variant over AWP (-0.041 [-0.135, 0.050]). AWP's graph had 2.23 times as many edges and took 2.17 times as long to solve, a consequence of its size.

Introduction

Decentralized finance (DeFi) runs lending, trading and asset management through smart contracts on public blockchains [1, 2]. Most of these services begin with an authorization: the owner of a token lets a contract move some of it. ERC-20 tokens store such an authorization as an allowance, set by calling `approve` or by signing an EIP-2612 `permit` [3, 4]. An allowance stays in force until it is spent, replaced or revoked, and most approvals on Ethereum are for unlimited amounts [5], so a malicious contract, or an honest one that is later compromised, can take the owner's entire balance of the approved token. Phishing sites lead users to sign exactly such approvals [6], and the

toolkits that do so are sold as a service [7]. Owners, their wallet software and protocols that build on other contracts therefore share one question: which spender contracts deserve an allowance.

Trust in a spender contract forms in a few ways today. Most owners decide when their wallet shows the approval request, and flaws in how wallets display and check approvals leave that moment open to fraud [8]. Code analysis judges a contract by its functions, for instance by flagging functions written for a rug pull [9]. Deployed reputation tools such as Human Passport weigh credentials to decide whether an address belongs to a distinct person [10], a question about the holder of a key that has no direct counterpart for a contract. None of these uses the public record of what other owners have decided, namely which contracts many owners have authorized.

Graph ranking offers a way to read that record. PageRank gives every node the stationary probability of a damped random walk along directed edges [11, 12], and EigenTrust and TrustRank adapt the walk to peer-to-peer trust and to web spam [13, 14]. On account-based ledgers the edges have so far been payments. Do and Do apply PageRank and HodgeRank to Ethereum transactions to measure social credit [15]. The Adaptive Weighted PageRank (AWP) of Do, Do and Nguyen gives each transfer a weight that rises with its value up to a ceiling and decays logistically with its age, and restarts the walk at addresses that sent recently [16]. Its authors check AWP by its agreement with the number and value of the transfers each address receives, statistics of the same transfers that make up its graph.

A transfer shows that value moved, not that the sender vouched for the recipient; routers, market makers, airdrop hunters and wash traders all produce transfers in bulk [17, 18]. An allowance records a different act. The holder of the funds names one spender and lets it move up to a stated amount. A new approval for the same owner, spender and token replaces the old one, so the latest allowance summarizes the relation, and approvals are rarer than transfers. An authorization need not express trust. To trade on an exchange a user must first approve its router, whatever the user thinks of it, and a phishing contract collects approvals as an honest one does. We therefore call the allowance edge an authorization and use the name EndorseRank only as the label of a method. Approvals have been used to cluster addresses of one entity [19], but we know of no reputation method that ranks contracts on them.

Whether either edge yields the better reputation score for contracts is partly a question of evaluation design, and two habits of the field make it hard to answer. One is to check a score against statistics of its own edges. Such agreement is expected by construction: it speaks to the coherence of the intended construct and not to what else the score can tell [20, 21]. The other is to compare walks only with other walks. A PageRank smooths a local count, and local counts are hard to beat. New links attach preferentially to nodes that already have many [22, 23], degree is a standard baseline in link prediction [24], also on Ethereum [25], and where degree correlations are weak PageRank is on average proportional to in-degree [26]. A propagated score earns its extra cost only where it beats the count underneath it.

This paper compares the two edge types as reputation scores for smart contracts on Arbitrum One. We score the same contracts with EndorseRank, PageRank on the latest allowances with AWP's edge weights and uniform restarts, with AWP and with the two raw in-degrees, freeze every score before two three-month label windows, and test each score on what happens afterwards. Every scored address holds contract code; externally owned accounts enter the graphs only as owners, senders and recipients. The design was fixed in steps that we report with their commits: an analysis plan before this study extracted any log, and a registration of the second window before this study extracted its logs. After the results of that registered analysis were known, we revised how amounts are valued, from raw token units over every ERC-20 token to US dollars over

five widely traded tokens. Results report the revised analysis. The rules fixed for the registered analysis are applied to it as a re-analysis, and the registered results are given beside it. There are three contributions.

- An out-of-window comparison of each walk with the raw degree it smooths, in two windows. Neither the plan nor the registration named these comparisons, and all were computed after the labels were known. Each edge type's scores rank later edges of the same type best. AWP ranked later new transfer senders above the transfer in-degree, whereas EndorseRank with the planned uniform restarts ranked later new approval pairs below the count of approving owners. A variant with AWP's activity restarts beat that count, but it was set against the count last, so the result is a hypothesis for a registered test.

- An exact expression for EndorseRank on spenders whose owners receive no approvals: a common constant times one plus the damped owner-normalized in-strength. It holds within 1% for every such spender, so for most spenders the walk is a reweighted approval count, while the remaining spenders show that multi-hop propagation is present in these graphs.

- A comparison of the two edge types on an outcome that neither graph contains, the later liquidation avoidance of contracts that trade on the perpetual futures exchange GMX V2, under rules registered before this study extracted the W1 outcomes (it had extracted the W0 closes but not scored them on liquidation), applied here to the revised scores as a re-analysis. In the primary window a lead of the approval count over the transfer count was not established, and the PageRank pair was not resolved; the samples are small.

A companion manuscript by the same authors (submitted to *IEEE Access*; unpublished) studies a walk that mixes the two edge types node by node, in the spirit of multilayer ranking [27, 28], its ablation, the two decision rules fixed for it in the plan and in the registration, and the Sybil exposure of the allowance walk. Both papers use the same extraction, cohorts, registration and revision of the amounts, and both report the spender holdout coefficients of the raw degrees, EndorseRank and AWP with their degree contrasts, the closed form of EndorseRank and the solver runtimes. The allowance-against-transfer comparison on trader outcomes (P, Q and rules R1–R3) is reported only here. Both papers report the revised analysis first, with the registered values beside them, and they report the same values for both analyses. On the same W1 trader cohort and liquidation-free close rate the companion reports only the contrasts of the mixed walk with its single-layer walks or AWP that its own registered rule names; here the two single-layer walks enter only through the registered pairs of the trader comparison.

Related work

On-chain reputation and risk. Besides the two PageRank studies above [15, 16], RiskProp rates Ethereum accounts by propagating risk through the transaction network [29], and network embeddings of the transaction graph help detect phishing addresses [30]. Surveys of blockchain trust and reputation systems find explicit ratings and transfer activity to be the usual signals [31, 32]. Decentralized credit scoring combines ledger records with prospect theory [33] or machine learning [34], and legal scholarship warns that such scores can become as opaque as the models they would replace [35]. A study of the ERC-20 transfer network finds that externally owned accounts and smart contracts follow different statistical patterns [36], one reason to

score contracts as a population of their own. These approaches score accounts by what they send and receive; none ranks contracts by the allowances that owners grant them.

Trust propagation and its weak points. HITS separates hubs from authorities [37]. EigenTrust normalizes local trust so that no peer can raise its own standing at will [13], and TrustRank lets trust flow outward from a vetted seed set [14]. Under uniform teleportation that protection is weaker: alliances of link farms lift a target's PageRank [38], and symmetric reputation functions such as PageRank cannot be Sybil-proof [39]. Both results apply to allowance graphs, where an approval costs a transaction fee and locks no tokens, and where whoever deploys a contract can create owner addresses at almost no cost [40].

Approvals. EIP-20 specifies the allowance, and EIP-2612 extends it to signed approvals [3, 4]. Wang et al. measure how common unlimited approvals are and how much value they expose [5]. Studies of transaction-based phishing and of drainer services show how attackers obtain approvals [6, 7], and Adamczyk dissects an approval-based fraud scheme and the wallet weaknesses it uses [8]. Victor's self-authorization heuristic reads approvals as links between addresses of one entity [19].

Baselines and evaluation. Degree is the natural benchmark for predicting new links [24, 25], and PageRank often behaves like a monotone function of in-degree [26]. Construct validity separates what a score measures from its agreement with checks built into it [20]. Writing an analysis down before its data are examined limits analytic flexibility [41]; where we could, we fixed each rule before this study extracted the data that judge it, and we state for each comparison when it was fixed.

Materials and methods

Data and cohorts

All events come from Arbitrum One, an optimistic rollup on Ethereum [42], read from the public BigQuery table `goog_blockchain_arbitrum_one_us.logs` (project `bigquery-public-data`). The observation window covers 33 months, 1 October 2023 through 30 June 2026 (the end, 23:59:59 UTC, is written T_{obs}). We read ERC-20 `Approval` and `Transfer` logs with exactly three topics, which leaves out the ERC-721 logs that share their signatures, and the `PositionDecrease` events of GMX V2, a perpetual futures exchange on Arbitrum One [43]. S1 Text gives the topics, decoded fields, label formulas and queries.

Contracts. We treat an address as a contract if `eth_getCode` at a pinned block returns non-empty code; an address with empty code, or with code that is only an EIP-7702 delegation designator [44], is an externally owned account (EOA). The analysis plan's candidate spenders, every address that received a non-zero approval from at least three distinct owners in the window, numbered 29,022. At block 513,216,506 (9 October 2026) 27,844 of them held contract code, 1,121 held none and 57 held a delegation designator. Only contracts are scored or evaluated. EOAs, which may belong to individuals, are present only as owners, senders and recipients and are never ranked or reported.

Tokens and prices. Amounts of different tokens can be added only after they are valued in one unit. We keep the approvals and transfers of five tokens, WETH, USDC, USDT0, ARB and WBTC, and value every amount in US dollars at DefiLlama's daily closing price [45]. We admitted a token if Chainlink's list of data feeds on Arbitrum One shows a market-price feed rated low or medium market risk for the token, for the asset it wraps one to one, or for the Ethereum token of which it is the canonical-bridge copy [46], and if the token occurs in the data. Of the 133 tokens that the list maps to Arbitrum One addresses, 64 qualified and 46 of these have a daily price; the five kept

are those with the most rows, together 75.75% of the rows of every token. Checked against Chainlink answers read on the chain for 38 dates per token, the DefiLlama prices were off by a median of 0.01% to 0.08% per token, with a largest gap of 0.72%. A transfer is valued at the price of its day and a latest allowance at the price of the anchor's day, because the edge represents the authorization that holds at the anchor. Keeping five tokens also removes the logs that spam and address-poisoning tokens can emit in the name of owners who never approved or paid anyone. Native ether moves without `Transfer` logs and enters the graphs only as WETH.

Cohorts. The matched cohort consists of the 15,107 contracts that at least three distinct owners approved, with a non-zero amount of one of the five tokens, during the observation window; they are the part of the plan's 27,844 contracts that meets the same rule on the five tokens. Only events with a cohort contract at one end were extracted: approvals in which it is owner or spender, and non-zero transfers in which it is sender or recipient. The graphs are therefore one-hop neighbourhoods of the cohort, and a path of two or more edges can exist only through cohort contracts that both grant and receive approvals. The analysis keeps 224,954,202 approval and 1,149,168,538 transfer logs of the observation window, 57.7% and 74.7% of the ego extraction of the plan's cohort. A spender cohort holds, at each freeze, every contract with a positive latest allowance of one of the five tokens in the extracted approvals: 17,050 contracts at the first freeze, of 17,167 such spenders, and 18,152 at the second. It includes contracts outside the matched cohort that a cohort contract had approved. A trader cohort holds the GMX V2 accounts with contract code that appear as a node of either graph at the freeze and close three or more positions in the label window: 88 contracts in the first window and 70 in the second. Over the observation window, contract accounts made 133,685 GMX V2 closes, including 6,473 liquidations; from July to September 2026 they made 6,805, including 182 liquidations. Few contract accounts traded in the label windows, so every trader result rests on few contracts. S1 Table summarizes windows, cohorts, graphs and amounts.

Windows. Window W0 freezes the scores at t_1 = 31 March 2026, 23:59:59 UTC, using every event since 1 October 2023, and counts its labels over April, May and June 2026. Window W1 freezes them at T_{obs}, so its scores are those of the same-window evaluation, and counts its labels over July, August and September 2026. The two windows share their score data up to t_1, and W1's score window contains W0's label window, so they are not independent samples.

Order of the analyses and what was fixed when

The comparisons were fixed at different times relative to the data, so we state the order of events (times in UTC; S1 File reproduces the three documents named).

- At 13:41:27 on 9 October 2026 an analysis plan and a versioned configuration were committed to the authors' public repository (commit `43d8045`); no log of this study had yet been extracted. Together they fixed the windows, the cohort and account-type rules, every score with its damping, decay, value transform and restart rule, EndorseRank's uniform restarts with the activity-restart variant to be reported beside it, the raw degrees, the same-window checks, the W0 labels and the statistics. Amounts were to stay in raw token base units over every ERC-20 token. The plan named no comparison of a PageRank with its raw degree.

- The observation window was extracted, and the same-window and W0 results of this registered analysis were computed.

- At 16:49:32 the registration of W1 was committed (commit `0d3a057`), once the W0 results were in, and before this study extracted any July–September log. Its

section on outcomes built from neither edge type fixed the trader cohort and labels, four pairs of an allowance-side and a transfer-side score, the decision contrasts P and Q, the stratification, the recipient comparison, the influence check and rules R1–R3 with the wording of each conclusion. The extraction script refuses to run without the committed registration; it recorded the registration's SHA-256 hash and started at 16:49:33, and it created the query for the July–September logs at 16:49:40, eight seconds after the commit. At that point no score of this study had been set against a liquidation label of the traders in either window, but an exploratory run had scored the W0 trader contracts on the four profit labels.

- The July–September logs were extracted and the registered evaluation was run. The comparisons of EndorseRank's activity variant with the approval count, and of EndorseRank with AWP in W1, were computed afterwards. The results of the registered analysis, with these later comparisons, were committed in `8d8b568`.

- On 10 October 2026, after all registered results were known, the valuation of amounts was revised. In raw base units AWP's value weight was within 10^{-4} of one for almost every amount, so the amount played no part in EndorseRank or AWP; the companion's amount-weighted layers added base units of tokens whose scales differ by twelve orders of magnitude; and every ERC-20 contract counted, including tokens that emit logs in the name of owners who took no part. The revision introduced the five tokens, the valuation in US dollars, the slopes of the value weight and a cap on allowances in the amount-weighted allowance layer. It was committed in three steps, at 07:54:35 (`a897475`), 08:12:44 (`09dfc32`) and 11:24:01 (`2705b28`), before any revised score was computed. The revised scores were computed from 11:36 and committed in `77e71f5`.

The revised analysis is the one reported in Results. Its scores were defined after the registered results were known, so no revised result counts as a registered one: rules R1–R3 are applied to it with their registered thresholds and wording, as a re-analysis, and the Results subsection "The registered analysis" sets the registered values beside the revised ones. Every comparison of a PageRank with its raw degree, in both analyses, was computed after the labels of its window were known. The closed form was derived to account for the computed scores and tests nothing.

Scores

Weighted PageRank

Every score is computed by one weighted PageRank solver. With edge weights $w(u, v)$, out-strength $o(u) = \sum_v w(u, v)$ and a restart distribution $\boldsymbol{\pi}$ over the node set V, the iteration is

$$\mathbf{r}^{(t+1)} = (1 - d)\,\boldsymbol{\pi} + d\,P^{\top}\mathbf{r}^{(t)} + d\Big(\sum_{u:\,o(u)=0} r_u^{(t)}\Big)\boldsymbol{\pi}, \qquad P_{uv} = \frac{w(u,v)}{o(u)} \text{ if } o(u) > 0, \quad (1)$$

where the rows of dangling nodes ($o(u) = 0$) are zero, so that the last term returns their mass through $\boldsymbol{\pi}$. The damping factor is $d = 0.85$; the iteration starts from the uniform vector and stops once the L_1 change falls below 10^{-8} or after 300 iterations. A cohort contract that a graph does not contain scores zero under that graph's methods; keeping it as an isolated node instead was fixed in the plan as a sensitivity analysis.

AWP on transfers

Following Do, Do and Nguyen [16], a transfer e worth x_e US dollars, made Δt_e days before the anchor, adds $\sigma(\Delta t_e)\,V(x_e)$ to the edge from its sender to its recipient, with

$$\sigma(\Delta t) = \frac{1}{1+\exp\big(k(\Delta t - t_0)\big)}, \qquad V(z) = \frac{2}{1+e^{-bz}} - 1, \qquad b = \frac{\ln 3}{m}, \tag{2}$$

so that

$$w_T(u,v) = \sum_{e:\,u\to v} \sigma(\Delta t_e)\,V(x_e), \qquad \pi_u \propto X_u = \sum_{v\neq u} \max_{e:\,u\to v} \sigma(\Delta t_e). \tag{3}$$

The restart weight X_u, which the authors call activeness, sums the time weight of u's latest transfer to each address it has paid, so an address that never pays another receives no restart mass. The paper gives no parameter values. The plan fixed $k = 0.01$ and $t_0 = 180$ days, so the time weight falls from $\sigma(0) = 0.858$ through $\sigma(90) = 0.711$ and $\sigma(182) = 0.495$ to $\sigma(365) = 0.136$. The revision set the slope from the median amount: $m_T = 200$ dollars, the median transfer up to t_1 rounded to two significant digits, so a 200-dollar transfer has value weight one half, a 400-dollar transfer 0.8 and a 100-dollar transfer 0.27. The anchor of the decay is the freeze date of each window.

EndorseRank on latest allowances

EndorseRank puts AWP's weights on the allowance graph. For each token, owner and spender only the most recent `Approval` up to the anchor is kept, in block and log order, and a latest value of zero deletes the entry. An edge $u \to v$ means that owner u holds a positive latest allowance for spender v on at least one token, and its weight is

$$w_A(u,v) = \sum_{\text{token}} \sigma\big(\Delta t^\star(u,v,\text{token})\big)\,V\big(a^\star(u,v,\text{token})\big), \tag{4}$$

where $a^\star$ is the latest allowance in US dollars and $\Delta t^\star$ the age of the approval that set it. The slope uses $m_A = 62$ dollars, the median finite latest allowance at t_1, so an allowance of a few hundred dollars weighs nearly as much as an unlimited one, and no allowance weighs more than one. EndorseRank restarts uniformly. Carrying AWP's rule over to allowances would send the walk back to owners in proportion to how many spenders each has approved, discounted by the age of each latest approval; that rewards the granting of approvals, while EndorseRank is meant to rank the contracts that receive them. The plan therefore fixed uniform restarts and named the variant with activity restarts as a score to report beside EndorseRank. The weight describes the latest approval, its amount and its time, and not what remains of the allowance at the anchor. An approval set before 1 October 2023 and never changed is invisible; spending through `transferFrom` does not lower the recorded amount of a finite allowance; and an owner that approves an allowance manager such as Uniswap's Permit2 is linked to the manager, not to the contracts the manager authorizes in turn (S1 Text).

Raw degrees and amount-weighted layers

The in-approve degree of a contract is the number of distinct owners that hold a positive latest allowance for it, and its transfer in-degree is the number of distinct other addresses that have sent it a transfer, both counted at the freeze:

$$k_A(v) = \big|\{u : a^\star(u,v,\cdot) > 0\}\big|, \qquad k_T(v) = \big|\{u \neq v : u \text{ sent a transfer to } v\}\big|. \tag{5}$$

Each count is the local statistic that one of the two PageRanks smooths, and out of window each serves as the baseline that its PageRank has to beat. The registered trader

comparison also sets against each other the two single-layer end points of the companion's mixed walk, written C-PR(λ=1) and C-PR(λ=0). Each walks one edge type with uniform restarts and with amounts in dollars instead of the bounded weight V: the allowance layer sums $\min(a^\star, U)$ over tokens, with $U = 20{,}000$ dollars, the 99th percentile of single transfers up to t_1, and the transfer layer sums $x_e\,\sigma(\Delta t_e)$ over transfers.

Checks and labels

Same-window checks. Each score is compared on the matched cohort with three families of local statistics computed over the observation window [16]. The transfer family holds a contract's in-degree and the summed dollar value of its inbound transfers. The allowance family holds its in-approve degree and the summed capped value of the allowances it holds. The Sybil-stability family holds the ratio of distinct inbound counterparties to inbound transfers, the span between its first and last transfer and its number of active months. The first two families come from the same events as the two graphs, so a score's agreement with the family of its own edges is an intended-construct check, partly built in, and not evidence of predictive power.

Spender labels. Two counts are taken in each label window, one per construct. New approval pairs counts the owners that give the spender a positive approval in the window while their pair held no positive latest allowance at the freeze. New transfer senders counts the distinct addresses that send the spender a transfer in the window without having sent it one between 1 October 2023 and the freeze, self-transfers excluded. Both count events of the five tokens only, and both count edges of the kinds that build the graphs, so agreement with them shows persistence of a construct, not anticipation of a different outcome. Both labels are sparse. Only 2,070 of the 17,050 W0 spenders (12.1%) gained any new approval pair over the three months, and 1,695 (9.9%) any new sender.

Trader labels. Six outcomes are computed from each trader contract's GMX V2 closes in the label window (formulas in S1 Text). A close is one `PositionDecrease` event, so partial closes count, and it counts as a liquidation if the event carries the liquidation order type or was emitted by the exchange's liquidation handler. The liquidation-free close rate (LFCR), the primary label, is one minus the share of closes that ended in liquidation, and the no-liquidation flag is one when none did. The number of profitable closes and the realized gain, the sum of positive profits, grow with the amount of trading; the profitable share and the non-loss share of closes do not. A liquidated close counts as neither profitable nor losing, so it lowers the profitable share but raises the non-loss share. The six outcomes come from the exchange's position events and not from the edges of either graph, and in that sense we call them outcomes built from neither edge type. They are not wholly separate from the transfer layer: the exchange settles collateral and profits in ERC-20 tokens, so earlier payouts can appear as transfers in the pre-freeze graph. A liquidation closes a leveraged position whose collateral no longer covers its losses [47]. It is not a credit default, since the position is settled against the trader's own collateral, but among the outcomes in these data it comes nearest to a counterparty failing to cover its obligations.

Comparison on outcomes built from neither edge type

The registration of W1 sets allowance-side scores against transfer-side scores in four pairs: the in-approve degree against the transfer in-degree; EndorseRank with activity restarts against AWP, which share the form of their edge weights and their restart rule and differ in the edge type; EndorseRank against AWP; and C-PR(λ=1) against C-PR(λ=0). C-PR(λ=0) minus AWP is reported to isolate AWP's weights and restarts.

All six labels are reported. W1 is the primary window and W0 the secondary one. Two contrasts on LFCR in W1 decide: P (in-approve degree minus transfer in-degree) and Q (EndorseRank with activity restarts minus AWP), both judged on Bonferroni 97.5% intervals. Three rules were registered.

- R1, a count-level lead on liquidation avoidance, holds if the 97.5% interval of P lies above zero in W1, the 95% interval of P stratified by the number of closes lies above zero in W1, and P is positive in W0.
- R2, a PageRank-level lead, holds if the 97.5% interval of Q lies above zero in W1 and Q is positive in W0.
- R3 applies if, in W1, P on the profitable share or on the non-loss share has a 95% interval entirely below zero; the allowance signal is then taken to be specific to avoiding liquidation.

The registration also fixed the wording of each conclusion. If R1 fails, there is "no count-level lead of allowance over transfer on later liquidation avoidance"; if R2 fails, "the PageRank versions are not resolved"; and if R3 applies, "the signal concerns forced-liquidation avoidance, not trading success". Traders that held at least one approval at the freeze, which we call recipients, are set against all other traders on every label, and as an influence check P is recomputed without the ten recipients with the most approving owners. The evaluation code committed with the registration also computes EndorseRank minus the in-approve degree and the coefficient of every single score, which the registration document does not name.

Statistical analysis

Agreement between a score and a check or label is measured by Kendall's rank correlation [48] in its tie-corrected form τ_b, on raw scores and raw values, so that tied scores stay tied. We write $\tau(a)$ for the coefficient of score a and $\Delta\tau = \tau(a) - \tau(b)$ for a paired contrast. Uncertainty comes from a paired bootstrap over contracts [49] with percentile intervals [50]: the evaluated cohort is resampled with replacement, and one matrix of resample indices is shared by every score, so a contrast keeps the covariance of its two coefficients. The same window and the spender holdouts use 400 resamples with seed 42, and the trader comparison 2,000 with the same seed. With 400 resamples each interval end carries a Monte Carlo error of about 0.13 bootstrap standard errors. Only P and Q in W1 are judged on Bonferroni 97.5% intervals; every other interval is unadjusted. A contrast is described as a difference when its interval excludes zero. The intervals treat contracts as independent and the graphs, and so the scores, as fixed, although spenders share owners. No p-values are reported, and no causal claim is made.

Two properties of the data bear on the comparisons. The raw degrees are integer counts with many ties, whereas EndorseRank leaves few (Results), so part of a gap between a count and a walk may reflect how τ_b treats ties rather than information. And a trader with more closes has more chances to be liquidated at least once, so for the traders τ_b is also calculated within strata of 3–4, 5–9, 10–24 and at least 25 closes in the label window and averaged with weights proportional to the number of contract pairs per stratum. The number of closes is measured after the freeze, so the stratification is descriptive.

Closed-form check and runtime

The closed form derived in Results was checked against EndorseRank as computed on the allowance graphs at both freezes: for every spender of the checked set the pipeline

compares the computed score with the closed form and records whether it lies within 1%. Runtime covers one solver call on prebuilt edge lists, with matrix assembly and power iteration: one untimed warm-up and five timed runs, under identical settings, of one Python implementation (SciPy sparse matrices, NumPy power iteration) on a machine with an Intel Core Ultra 9 185H processor and 64 GB of memory under Windows 11 (Python 3.11.9, NumPy 2.3.2, SciPy 1.16.1).

Ethics statement

This study used only publicly available blockchain data and public token prices and involved no human participants. The scored addresses are contracts; no externally owned account is scored or reported, no attempt was made to link any address to a real-world identity, and only aggregate statistics are reported. [ethics clearance reference to be added by the authors]

Data availability

The evaluation code, versioned configuration, SQL queries, analysis plan, W1 registration, revision document, extraction manifest, the machine-readable summaries behind every table with their SHA-256 checksums, and the per-contract scores and labels behind every coefficient are publicly available in the Git repository `https://github.com/abitocodes/phd_works`, folder `1-Dissertation-Works/ERC20-Allowance-PageRank-Contract-Reputation/experiments`, at these commits of its default branch: plan `43d8045`, registration `0d3a057`, registered results `8d8b568`, revision `a897475`, `09dfc32` and `2705b28`, revised results `77e71f5`. The folder as published with the revised results is at `https://github.com/abitocodes/phd_works/tree/77e71f5503afe57aa2f653b875ad9ace46e8b132/1-Dissertation-Works/ERC20-Allowance-PageRank-Contract-Reputation/experiments`. The summaries are under `data/3-published-results-for-thesis/` and the per-contract files under `data/2-processed-tables-and-evaluations/`. The scripts that check the closed-form and trader values against those files are in `https://github.com/abitocodes/phd-dissertation-draft-en`, folder `analysis/`, and the scripts that generate the tables and Fig 1 of this article are in its folder `manuscripts/A-plos-one/`. The pre-specification documents are also S1 File, and the queries are in S1 Text. The public BigQuery table is not a versioned snapshot, so the extracted graph tables were kept in the authors' BigQuery project, and the repository holds their manifests; the BigQuery jobs of the whole study billed 5.57 TiB and cost USD 34.80. The per-contract files hold addresses only as they appear on the public ledger, with no off-chain identity.

Use of generative AI

Claude (Anthropic), used through the Claude Code agent, was used in this study for more than language editing. Under the first author's direction it wrote and ran parts of the extraction and evaluation code in the first repository. In the second repository it wrote the scripts of the folder `analysis/`, which copy the closed-form and trader values from the published summaries and recompute the trader values from the per-contract files as a check, and the scripts that generate the tables, Fig 1 and S1 and S2 Tables of this article. It also drafted and revised the text of this manuscript, its supporting information and the cover letter, and checked text and scripts for errors. Every number reported here was checked against the published summaries with scripted checks. The authors take full responsibility for the content of the article.

Results

Graph structure and coverage

On the same contracts the two graphs differ in size and shape (Table 1). The allowance graph at T_{obs} holds 11,210,114 edges among 6,221,665 nodes; the transfer graph holds 25,001,216 edges among 7,884,602 nodes, 2.23 times as many edges. A revoked allowance removes its edge, while any pair that exchanged a transfer at some point in the window keeps a transfer edge. Owners holding a positive allowance account for 6,208,769 of the allowance graph's nodes, so that graph is a large fan of owners pointing at comparatively few spenders, and 5,763,757 of its 12,590,613 positive latest allowances (45.8%) are unlimited. Coverage of the cohort also differs. Of the 15,107 cohort contracts, 523 lie outside the allowance graph, because every allowance they had received was later set to zero and they hold none as owners; 2,670 lie outside the transfer graph, and 80 outside both. EndorseRank leaves few ties among the cohort (13,625 distinct scores; 0.17% of contract pairs tied) and AWP somewhat more (12,435 distinct scores; 3.13% of pairs tied), most of them among the contracts that score zero.

Table 1. Graphs at the two freeze dates and fit of the closed form for EndorseRank.

	T_{obs}, 30 Jun 2026 (same window and W1)	t_1, 31 Mar 2026 (W0)
Allowance graph, nodes	6,221,665	–
Allowance graph, edges	11,210,114	10,650,258[a]
Transfer graph, nodes	7,884,602	–
Transfer graph, edges	25,001,216	23,680,692[a]
Contracts checked	15,107 (matched cohort)	17,050 (W0 spenders)
Spenders among them (allowance in-edge)	14,406	17,032
depth one (no owner has an in-edge)	12,039	11,326
deeper (an owner is itself approved)	2,367	5,706
Common score c of nodes without in-edges	7.64×10^{-8}	8.14×10^{-8}
Within 1% of the closed form[b]: depth-one spenders	100.0%	100.0%
Within 1% of the closed form[b]: deeper spenders	26.4%	21.4%
Within 1% of the closed form[b]: all spenders	87.9%	73.7%

Graphs built from the ego events of the matched cohort up to each freeze date (revised analysis: approvals and transfers of WETH, USDC, USDT0, ARB and WBTC, amounts in US dollars). At T_{obs} the edge counts are those of pairs with a positive EndorseRank or AWP weight. A spender is a contract of the checked set with an allowance in-edge of positive weight; it has depth one when none of its owners has an in-edge itself. Damping $d = 0.85$. [a]Owner–spender and sender–recipient pairs in the pair tables at t_1, counted before pairs whose value weight is zero in floating point are dropped; the node counts at t_1 are not recorded. [b]Eq (7), derived in Results (What EndorseRank computes on depth-one spenders). Sources: `same-window` and `holdout-w0/eval_summary.json`, keys `graph` and `closed_form`, through `analysis/closed-form/closed_form_check.json`.

Each score agrees most with the checks of its own edges

In the same window each score follows the edges it walks on (Table 2). On the allowance family EndorseRank reaches $\tau = 0.585$ $[0.578, 0.592]$ against 0.193 $[0.182, 0.203]$ for AWP; on the transfer family AWP reaches 0.572 $[0.564, 0.580]$ against 0.158 $[0.147, 0.169]$ for EndorseRank. The paired differences are $\Delta\tau = +0.392$ $[0.380, 0.405]$ (EndorseRank minus AWP, allowance family) and $+0.414$ $[0.401, 0.427]$ (AWP minus EndorseRank, transfer family). This is the pattern an intended-construct check expects, and part of it is built in: EndorseRank agrees most with the in-approve degree (0.655 $[0.648, 0.662]$), the count it smooths. With dollar amounts, AWP's agreement with the value of inbound transfers (0.566 $[0.558, 0.575]$) comes close to its agreement with the number of distinct senders (0.578 $[0.569, 0.587]$), as its value weight

intends. On the Sybil-stability family, which averages transfer-based measures of how long and how evenly a contract has received transfers, AWP reaches 0.395 [0.387, 0.403] and EndorseRank 0.007 [−0.001, 0.015], an interval that includes zero. With activity restarts EndorseRank agrees less with its own family (0.426 [0.418, 0.435]), because its restarts then land mostly on the owners that grant the most approvals.

Table 2. Same-window agreement of each score with the checks of each edge type (intended-construct check).

Check (Kendall τ_b [95% CI])	EndorseRank	EndorseRank, activity restarts	AWP
Transfer family (mean)	0.158 [0.147, 0.169]	0.164 [0.153, 0.173]	0.572 [0.564, 0.580]
in-degree	0.182 [0.169, 0.194]	0.179 [0.167, 0.190]	0.578 [0.569, 0.587]
in-value (USD)	0.134 [0.123, 0.145]	0.149 [0.137, 0.158]	0.566 [0.558, 0.575]
Allowance family (mean)	0.585 [0.578, 0.592]	0.426 [0.418, 0.435]	0.193 [0.182, 0.203]
in-approve degree	0.655 [0.648, 0.662]	0.447 [0.438, 0.457]	0.206 [0.193, 0.217]
in-approve value (USD, capped)	0.515 [0.506, 0.524]	0.406 [0.396, 0.415]	0.180 [0.169, 0.191]
Sybil-stability family (mean)	0.007 [−0.001, 0.015]	0.049 [0.039, 0.057]	0.395 [0.387, 0.403]
Paired differences $\Delta\tau$ [95% CI] and agreement of the two rankings			
EndorseRank − AWP, allowance family	+0.392 [0.380, 0.405]		
AWP − EndorseRank, transfer family	+0.414 [0.401, 0.427]		
τ_b between EndorseRank and AWP	0.174 [0.163, 0.185]		
with missing contracts as isolated nodes	0.174 [0.163, 0.185]		

Matched cohort, $n = 15{,}107$ contracts, scores at T_{obs} (30 June 2026). Each check is computed from the events of one graph: the transfer family from inbound transfers, the allowance family from latest allowances received. Agreement of a score with the checks of its own edge type is therefore partly built in. The Sybil-stability family averages the inbound-counterparty ratio, transfer tenure and active months. Family means are plain averages of the coefficients of their checks. Paired contract bootstrap, 400 resamples (seed 42); intervals are unadjusted. None of these comparisons decides a rule. Source: `revised-usd/same-window/eval_summary.json`, keys `families`, `per_proxy`, `contrasts`, `inter_method`.

The two walks order the contracts differently. Their rankings agree at $\tau_b = 0.174$ [0.163, 0.185]: both scores order 96.7% of the 114,103,171 pairs of cohort contracts, and they disagree on the direction for 41.2% of those. Keeping the contracts outside a graph as isolated nodes instead of scoring them zero changes this agreement at none of the three reported decimals. A protocol that relies on a transfer-based score cannot assume that its ordering of contracts survives a switch to an allowance-based one.

Out of window, each edge type ranks later edges of its own type

Table 3 sets every score frozen at each date against the two spender labels. As persistence would lead one to expect, the edge type decides which label a score ranks best. In W0 the in-approve degree ranks new approval pairs at $\tau = 0.250$ [0.237, 0.262], EndorseRank at 0.231 [0.220, 0.242] and its activity variant at 0.308 [0.299, 0.319], while AWP (0.196 [0.181, 0.210]) and the transfer in-degree (0.135 [0.121, 0.150]) stay lower. On new transfer senders the order reverses: AWP reaches 0.346 [0.336, 0.356] and the transfer in-degree 0.303 [0.292, 0.315], against 0.200 [0.189, 0.210] for EndorseRank. EndorseRank minus AWP is $\Delta\tau = +0.035$ [0.019, 0.051] on new approval pairs and −0.146 [−0.156, −0.136] on new senders. W1 repeats the pattern (Table 3): EndorseRank minus AWP is +0.065 [0.051, 0.081] on new approval pairs and −0.136 [−0.146, −0.127] on new senders.

AWP beats the transfer count; EndorseRank does not beat the approval count

Against the raw degree it smooths, on the label of its own edge type, each walk behaves differently (Fig 1A). AWP ranks later new transfer senders above the transfer in-degree

Table 3. Spender holdouts: scores frozen at each date against labels counted over the next three months.

	W0: freeze 31 Mar 2026 ($n = 17{,}050$)		W1: freeze 30 Jun 2026 ($n = 18{,}152$)	
Score at the freeze	New approval pairs	New senders	New approval pairs	New senders
In-approve degree (count)	0.250 [0.237, 0.262]	0.180 [0.169, 0.192]	0.268 [0.257, 0.279]	0.205 [0.193, 0.215]
Transfer in-degree (count)	0.135 [0.121, 0.150]	0.303 [0.292, 0.315]	0.131 [0.119, 0.143]	0.308 [0.299, 0.319]
EndorseRank	0.231 [0.220, 0.242]	0.200 [0.189, 0.210]	0.234 [0.222, 0.244]	0.194 [0.184, 0.205]
EndorseRank, activity restarts	0.308 [0.299, 0.319]	0.269 [0.260, 0.278]	0.295 [0.285, 0.304]	0.242 [0.232, 0.253]
AWP	0.196 [0.181, 0.210]	0.346 [0.336, 0.356]	0.169 [0.157, 0.183]	0.330 [0.321, 0.340]
Each PageRank against the raw degree it smooths, on the label of its own edge type: $\Delta\tau$ [95% CI]				
EndorseRank − in-approve degree	−0.019 [−0.026, −0.011]		−0.034 [−0.038, −0.029]	
EndorseRank, activity restarts − in-approve degree	+0.059 [0.050, 0.067]		+0.027 [0.021, 0.034]	
AWP − transfer in-degree		+0.043 [0.038, 0.048]		+0.022 [0.017, 0.027]
Allowance walk against transfer walk: $\Delta\tau$ [95% CI]				
EndorseRank − AWP	+0.035 [0.019, 0.051]	−0.146 [−0.156, −0.136]	+0.065 [0.051, 0.081]	−0.136 [−0.146, −0.127]

Kendall τ_b with 95% percentile interval from 400 paired contract resamples (seed 42); intervals are unadjusted. Spenders are the contracts with a positive latest allowance of one of the five tokens at the freeze. In W0, 2,070 spenders gained at least one new approval pair and 1,695 at least one new transfer sender; in W1, 1,616 and 1,533 did. Every paired difference in this table was computed after the labels of its window were known, and none of them decides a rule; the variant with activity restarts was set against the in-approve degree last. Sources: `revised-usd/holdout-w0/eval_summary.json` (`taus`, `contrasts_after`), `window-w1/eval_summary.json` (`spender_taus`, `endorserank_results`), `supplementary/supplementary.json`.

in both windows, by $\Delta\tau = +0.043$ $[0.038, 0.048]$ in W0 and $+0.022$ $[0.017, 0.027]$ in W1. So on the transfer side AWP's walk, with its time and value weights and its restarts at active senders, ranked these arrivals above the count. EndorseRank, restarting uniformly as the plan fixed, ranks later new approval pairs below the in-approve degree in both windows, by -0.019 $[-0.026, -0.011]$ in W0 and -0.034 $[-0.038, -0.029]$ in W1. On the allowance side the count of approving owners was the better predictor. Neither the plan nor the registration document named any of these contrasts. In the registered analysis the evaluation code committed with the registration, before this study extracted the W1 logs, already contained the two W1 contrasts of EndorseRank and AWP with their degrees; the W0 contrasts, the activity-variant contrasts and every contrast of the revised analysis were chosen and computed after the labels were known.

The activity-restart variant of EndorseRank beats the in-approve degree on new approval pairs in both windows, by $+0.059$ $[0.050, 0.067]$ in W0 and $+0.027$ $[0.021, 0.034]$ in W1, and EndorseRank itself by $+0.077$ $[0.071, 0.084]$ and $+0.061$ $[0.054, 0.067]$. The two variants differ in nothing but where the walk restarts, so on the allowance side it seems to be the restart vector that decides whether the walk improves on the count. Because these comparisons were made after the labels of each window were known, and the variant was set against the count last, the result is a hypothesis for a registered test and not a finding. The variant's lower same-window agreement with its own family (Table 2) also shows that same-window agreement does not rank scores of one construct by what they anticipate.

Coverage does not explain the degree comparisons. AWP gives zero to the 2,891 W0 spenders that had no transfer edge before the freeze. These spenders went on to gain new approval pairs at about the rate of the others (13.3% against 11.9%) but hardly ever gained a new sender (0.6% against 11.9%). Restricted to the 14,094 spenders with an inbound transfer from another address before the freeze, a subset chosen after the results were known, both degree comparisons keep their signs: AWP minus the transfer in-degree is $+0.056$ $[0.049, 0.064]$ on new senders, and EndorseRank minus the in-approve degree is -0.013 $[-0.020, -0.006]$ on new approval pairs. Ties may play a part on both sides. The in-approve degree and the transfer in-degree are heavily tied integer counts, whereas EndorseRank gives 16,208 distinct scores to the 17,050 W0 spenders and ties 0.027% of their pairs, and AWP gives 2,891 of them the same score of

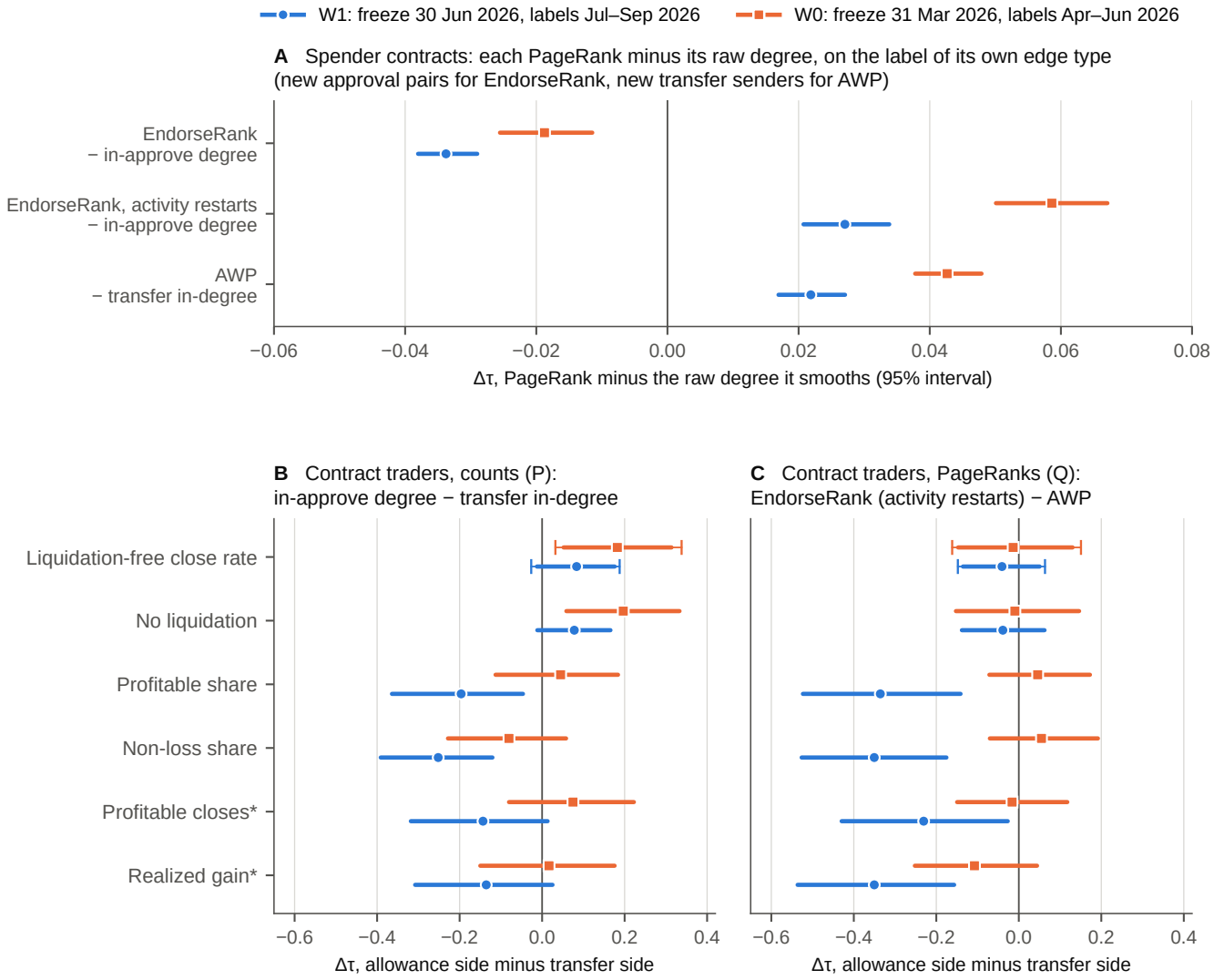


Fig 1. PageRanks against their raw degrees on spender contracts, and allowance side against transfer side on contract traders. Each point is a paired difference in Kendall's τ_b between two scores frozen at the start of a three-month label window. Blue circles: W1, freeze 30 June 2026, labels July–September 2026. Orange squares: W0, freeze 31 March 2026, labels April–June 2026. Thick lines are unadjusted 95% paired bootstrap intervals. (A) Spender contracts ($n = 17{,}050$ in W0 and 18,152 in W1; 400 resamples): each PageRank minus the raw degree it smooths, on new approval pairs for the two EndorseRank variants and on new transfer senders for AWP; computed after the labels were known. (B) Contract traders ($n = 88$ in W0 and 70 in W1; 2,000 resamples): in-approve degree minus transfer in-degree (contrast P) on six later outcomes. (C) The same for EndorseRank with activity restarts minus AWP (contrast Q). On the first row of (B) and (C), thin lines with end ticks show the 97.5% intervals; in W1 these are the decision intervals of the registered rules, applied here to the revised scores. An asterisk marks the two outcomes that grow with the number of closes. Revised analysis (amounts in US dollars, five tokens).

zero and ties 2.87% of their pairs; τ_b treats these differently, and no tie-robust comparison was made.

Two further labels, which the plan did not name, were computed on the W0 spenders: revocations, the owners of a positive allowance at the freeze that set it to zero in the label window, and drained owners, a heuristic count of owners whose new approval was followed within hours by an outgoing transfer of the same token (S1 Text). Every score ranks both positively. Revocations follow the in-approve degree most closely (0.388 [0.376, 0.398]) and AWP least (0.191 [0.175, 0.205]); for drained owners the coefficients run from 0.112 [0.096, 0.126] for the transfer in-degree to 0.328 [0.318, 0.338] for EndorseRank with activity restarts. A contract that owners trusted more should lose approvals less often, not more, so these counts fit the volume of owner activity rather than endorsement, and neither is offered as evidence.

What EndorseRank computes on depth-one spenders

The one-hop extraction makes most nodes of the allowance graph either pure owners or pure spenders. A cohort contract can be both, since it can receive approvals and approve other contracts, so the graph also has longer paths. For a spender whose owners receive no approvals, EndorseRank can be written out.

Observation. Let PageRank follow Eq (1) with damping d and uniform restarts, $\pi_u = 1/|V|$, and let r_D be the total score of the dangling nodes. Every node without in-edges then has the common score

$$c = \frac{1 - d + d\, r_D}{|V|}, \tag{6}$$

and a node v none of whose in-neighbors has an in-edge of its own scores

$$r_v = c\,\big(1 + d\,\delta(v)\big), \qquad \delta(v) = \sum_u \frac{w(u,v)}{o(u)}. \tag{7}$$

Here $\delta(v)$ is the owner-normalized in-strength of v: each owner u contributes the share of its own approval weight that goes to v, so an owner that approves many spenders adds little to each of them.

Derivation. At the fixed point of Eq (1) every node v receives $(1-d)/|V|$ from teleportation and $d\, r_D/|V|$ from the dangling rows, together c, plus $d\sum_u r_u\, w(u,v)/o(u)$ from its in-neighbors. A node without in-edges receives only c. If every in-neighbor u of v is such a node, substituting $r_u = c$ gives Eq (7).

The expression is a one-step expansion of the fixed point rather than a closed form in the strict sense, because c depends on r_D, which the whole score vector determines; but c is common to all spenders it covers, so EndorseRank orders those spenders exactly as $\delta(v)$ does. It describes the computed scores (Table 1). In the graph at T_{obs} every node without in-edges has the same score, $c = 7.64 \times 10^{-8}$, up to floating-point error. Among the 14,406 cohort spenders, the 12,039 of depth one all lie within 1% of Eq (7); among the remaining 2,367, whose owners include at least one approved address, only 26.4% do, which makes 87.9% over all spenders. On the graph at t_1, 11,326 of the 17,032 W0 spenders have depth one, again all within 1%, and 73.7% of all W0 spenders lie within 1%. The W0 cohort has more deeper spenders partly because it includes contracts outside the matched cohort, which enter the data only through approvals from cohort contracts.

This has two consequences. First, for a depth-one spender the walk does nothing but reweight its approving owners: an approval weighs more when it is recent and large, up to a weight of one, and less when its owner spreads its approvals over many spenders. Apart from these weights and this division, EndorseRank and the count carry the same

information about such a spender, and there is nothing further for propagation to supply. We did not isolate which of the weights costs EndorseRank its agreement with later approvals. Second, longer paths exist. Roughly one cohort spender in six at T_{obs}, and one W0 spender in three, is approved by an owner that others approve in turn, and most of these scores differ from the closed form by more than 1%. Propagation over several steps therefore occurs in these graphs; it did not, however, lift EndorseRank with uniform restarts above the raw count out of window. AWP, which walks a transfer graph with cycles, has no comparable closed form.

Outcomes built from neither edge type

Every result in this subsection applies the registered pairs, rules and wording to the revised scores, after the registered results were known, and is a re-analysis; Table 4 gives the contrasts on LFCR and Fig 1B and 1C all six labels. With 70 contract traders in W1 and 88 in W0, all intervals here are wide. Of the 70 W1 traders, 9 had at least one liquidation and 1 had every close liquidated.

Table 4. Allowance-side minus transfer-side scores on the later liquidation-free close rate of contract traders.

Contrast ($a - b$)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI	97.5% CI
W1: freeze 30 June 2026, labels July–September 2026 ($n = 70$ contract traders)					
P: in-approve degree − transfer in-degree	0.394	0.310	+0.084	[−0.013, 0.176]	[−0.026, 0.188]
P, stratified by closes	0.381	0.280	+0.101	[−0.024, 0.231]	–
Q: EndorseRank (activity restarts) − AWP	0.288	0.329	−0.041	[−0.135, 0.050]	[−0.148, 0.064]
Q, stratified by closes	0.273	0.301	−0.028	[−0.207, 0.098]	–
EndorseRank − AWP	0.297	0.329	−0.032	[−0.133, 0.066]	–
EndorseRank − in-approve degree[a]	0.297	0.394	−0.097	[−0.174, −0.009]	–
C-PR(λ=1) − C-PR(λ=0)[b]	0.301	0.350	−0.049	[−0.169, 0.060]	–
C-PR(λ=0) − AWP[b]	0.350	0.329	+0.021	[−0.016, 0.065]	–
W0: freeze 31 March 2026, labels April–June 2026 ($n = 88$ contract traders)					
P: in-approve degree − transfer in-degree	0.344	0.161	+0.182	[0.052, 0.314]	[0.032, 0.338]
P, stratified by closes	0.338	0.209	+0.129	[0.020, 0.293]	–
Q: EndorseRank (activity restarts) − AWP	0.227	0.240	−0.014	[−0.147, 0.130]	[−0.161, 0.151]
Q, stratified by closes	0.215	0.252	−0.037	[−0.173, 0.125]	–
EndorseRank − AWP	0.224	0.240	−0.017	[−0.156, 0.128]	–
EndorseRank − in-approve degree[a]	0.224	0.344	−0.120	[−0.203, −0.033]	–
C-PR(λ=1) − C-PR(λ=0)[b]	0.225	0.265	−0.041	[−0.187, 0.105]	–
C-PR(λ=0) − AWP[b]	0.265	0.240	+0.025	[−0.022, 0.077]	–

Kendall τ_b of score a and score b on the GMX V2 contract traders with at least three closes in the label window, and their difference. Paired contract bootstrap with 2,000 resamples (seed 42). The pairs, P and Q with their Bonferroni 97.5% intervals, the stratified rows and the rules R1–R3 were registered for W1 before this study extracted its logs; here they are applied to the revised scores, after the registered results were known, so no row is a registered test. Only the 97.5% intervals of P and Q in W1 enter a rule; every other interval is unadjusted. Stratified rows combine τ_b within strata of 3–4, 5–9, 10–24 and 25 or more label-window closes, weighted by contract pairs. [a]Not named in the registration document; computed by the code committed with it. [b]Amount-weighted allowance and transfer layers walked alone with uniform restarts. Source: `analysis/neutral-label/results.json`, `analyses.revised.windows`; S2 Table gives all six labels.

Counts (R1 does not hold). Among the W1 traders the highest in-approve degree is one, so P in effect compares the 37 traders that had received an approval with the 33 that had not. The in-approve degree ranks the later LFCR at $\tau = 0.394$ $[0.265, 0.518]$ and the transfer in-degree at 0.310 $[0.184, 0.425]$, so the transfer count also ranks this outcome. P is $+0.084$ $[-0.013, 0.176]$, with a 97.5% interval of $[-0.026, 0.188]$ that includes zero, and $+0.101$ $[-0.024, 0.231]$ stratified by the number of closes. W0 gives P = $+0.182$ (97.5% interval $[0.032, 0.338]$), which meets the W0 condition of R1, but the two W1 conditions fail. The registered wording for this case applies: there is no count-level lead of allowance over transfer on later liquidation avoidance.

PageRanks (R2 does not hold). Q, EndorseRank with activity restarts minus AWP, is -0.041 in W1 (97.5% interval $[-0.148, 0.064]$) and -0.014 in W0 (97.5% interval $[-0.161, 0.151]$). In the registered wording, the PageRank versions are not resolved. The

97.5% interval of Q in W1 is 0.21 wide, so at these sample sizes an estimate of 0.1 or less, the size of every degree contrast on the spender labels and of EndorseRank minus AWP on new approval pairs (Table 3), would not have been resolved. The other pairs are not resolved either: EndorseRank minus AWP is -0.032 $[-0.133, 0.066]$ in W1 and -0.017 $[-0.156, 0.128]$ in W0, and the amount-weighted layers differ by -0.049 $[-0.169, 0.060]$ and -0.041 $[-0.187, 0.105]$. In a contrast that the registration document did not name, EndorseRank falls below the approval count in both windows, by -0.097 $[-0.174, -0.009]$ in W1 and -0.120 $[-0.203, -0.033]$ in W0. Here, as on the spender labels, the walk with uniform restarts loses to the count it smooths.

The other labels (R3 applies). The two shares that do not grow with trading run the other way. In W1 the transfer count ranks traders above the approval count on both shares, with P at $\Delta\tau = -0.196$ $[-0.364, -0.046]$ for the profitable share and -0.252 $[-0.391, -0.120]$ for the non-loss share. R3 therefore applies, and the registered wording reads: the signal concerns forced-liquidation avoidance, not trading success. A liquidated close counts as not losing, so some of the gap on the non-loss share follows from the definition; the profitable share counts such a close as not profitable, and R3 rests mainly on that share. On the no-liquidation flag P is $+0.078$ $[-0.012, 0.166]$ in W1 and $+0.197$ $[0.059, 0.333]$ in W0. On the two outcomes that grow with trading, the PageRank pairs favor the transfer side in W1: Q is -0.231 $[-0.429, -0.027]$ on profitable closes and -0.350 $[-0.536, -0.157]$ on realized gain, where AWP reaches 0.346 $[0.189, 0.490]$. In W1 the same pairs also favor the transfer side on both shares (S2 Table). In W0 no allowance-against-transfer pair is resolved on either share, and on the four labels of trading success only the amount-weighted layers on realized gain are (-0.248 $[-0.433, -0.067]$). EndorseRank minus the in-approve degree, a contrast the registration document did not name, is $+0.100$ $[0.009, 0.194]$ on the W0 non-loss share.

Who the recipients are. None of the recipients was liquidated in either window (37 of 37 in W1, 40 of 40 in W0), whereas 72.7% and 75.0% of the other traders went without a liquidation (Table 5). Recipients were not simply less active: they closed more positions (median 24 against 11 in W1 and 31 against 10 in W0) and had lower mean profitable and non-loss shares (0.325 against 0.479 and 0.433 against 0.644 in W1), the pattern behind R3. Dropping the ten recipients with the most approving owners leaves P at $+0.073$ $[-0.039, 0.186]$ in W1 and $+0.188$ $[0.046, 0.331]$ in W0; with nearly all degrees equal to one, the ten dropped are taken in node order. All traders in both windows are contracts, so the split into recipients and others is not a split by account type. One explanation would fit every row of Table 5: recipients may be vault or managed-trading contracts that others authorize to move their tokens, that trade at low leverage and adjust positions often, and so are rarely liquidated, close many partial positions and earn modest profitable shares. The data hold neither leverage nor the role of each contract, so this explanation is untested. How many recipients the two windows share was not counted.

The registered analysis

The registered analysis, with raw base units over every ERC-20 token and the plan's cohort of 27,844 contracts, reached the same verdicts (Table 6). There, P in W1 was $+0.063$ $[-0.026, 0.148]$ with a 97.5% interval of $[-0.040, 0.158]$, Q in W1 was $+0.051$ with a 97.5% interval of $[-0.104, 0.232]$, R1 and R2 did not hold, and R3 applied; these are its registered results. Its degree contrasts, which no rule decides, point the same way as the revised ones in both windows: EndorseRank fell below the in-approve degree, AWP rose above the transfer in-degree, and the activity variant rose above the in-approve degree, each with an interval that excludes zero. EndorseRank met the closed form for every depth-one spender there as well. The revision changed the coefficients, most visibly the same-window agreement of each score with its own family,

Table 5. Contract traders that had received an approval at the freeze (recipients) against all other traders.

	W1 (July–September 2026)		W0 (April–June 2026)	
	Recipients	Other traders	Recipients	Other traders
Traders	37	33	40	48
Share with no liquidation	1.000	0.727	1.000	0.750
Mean liquidation-free close rate	1.000	0.878	1.000	0.919
Mean profitable share	0.325	0.479	0.362	0.474
Mean non-loss share	0.433	0.644	0.402	0.607
Median closes in the label window	24.0	11.0	31.0	10.0
Highest in-approve degree at the freeze	1	0	3	0
P without the ten recipients of highest in-approve degree	+0.073 [−0.039, 0.186] ($n = 60$)		+0.188 [0.046, 0.331] ($n = 78$)	

Recipients have an in-approve degree above zero at the freeze. Every trader in both windows holds contract code. The comparison and the influence check (P on the liquidation-free close rate, 95% interval from 2,000 paired resamples) were registered with the contrasts of Table 4 and are applied here to the revised scores. With tied in-approve degrees, the ten recipients left out are taken in node order. Source: `analysis/neutral-label/results.json`, keys `recipients` and `influence_without_top10`.

but no verdict and no sign of a degree contrast. It did reverse the sign of every allowance-against-transfer PageRank pair on the liquidation-free close rate: Q, EndorseRank minus AWP and the amount-weighted layers were positive in both windows of the registered analysis and are negative in both windows of the revised one, all with intervals that include zero (Table 6).

Computational cost

With one solver and identical settings, EndorseRank's solver graph (11,210,114 edges on 6,219,063 nodes) took 10.29 s on average and AWP's (25,001,216 edges on 7,882,356 nodes) 22.37 s, over five timed runs with standard deviations of 0.60 and 1.83 s. AWP thus took 2.17 times as long at an edge ratio of 2.23, although it needed more iterations (92 against 69), and its peak memory was 1,712.6 MB against 879.7 MB. Matrix assembly and each power iteration are linear in the number of edges [51], so the difference in time follows from the sparser allowance graph and not from any property of the algorithm. Any walk that also uses the transfer edges pays for them: the companion's mixed walk over both edge sets (31,552,359 edges) took 36.16 s.

Discussion

The edge type sets which later arrivals a score ranks; EndorseRank did not beat the approval count

The two edge types measure different things, and the out-of-window results show it in both windows: scores built on allowances rank which contracts later gain new approvals, and scores built on transfers which later gain new senders. Choosing the edge is therefore a decision about what a reputation score for contracts is meant to follow. Whether propagating along the chosen edge adds anything depended on the side. In comparisons made after the labels were known, AWP beat the transfer count in both windows, and EndorseRank with uniform restarts fell below the approval count in both. That a count is hard to beat is unsurprising given preferential attachment and link-prediction practice [22, 24], but on-chain reputation work rarely tests it, comparing walks with each other or with statistics of their own edges [16].

The closed form says what EndorseRank computes for most spenders: the approval count, with each approval scaled by how recent and how large it is and then divided by

Table 6. Registered analysis beside the revised analysis.

	Registered analysis	Revised analysis (re-analysis)
Data and same window		
Amounts; tokens	raw base units; every ERC-20	US dollars; five tokens
Matched cohort (contracts)	27,844	15,107
Allowance / transfer edges at T_{obs}	22,321,470 / 59,737,617	11,210,114 / 25,001,216
EndorseRank, allowance family	0.481 [0.476, 0.487]	0.585 [0.578, 0.592]
AWP, transfer family	0.447 [0.440, 0.453]	0.572 [0.564, 0.580]
τ_b between EndorseRank and AWP	0.217 [0.209, 0.225]	0.174 [0.163, 0.185]
Depth-one / all spenders within 1% of Eq (7)	100.0% / 86.4%	100.0% / 87.9%
Spender holdouts, PageRank minus its raw degree (computed after the labels)		
Spender cohorts, W0 / W1	33,101 / 34,587	17,050 / 18,152
W0: EndorseRank − in-approve degree, new pairs	−0.011 [−0.016, −0.006]	−0.019 [−0.026, −0.011]
W0: EndorseRank (activity) − in-approve degree	+0.043 [0.036, 0.049]	+0.059 [0.050, 0.067]
W0: AWP − transfer in-degree, new senders	+0.038 [0.035, 0.042]	+0.043 [0.038, 0.048]
W1: EndorseRank − in-approve degree, new pairs	−0.029 [−0.032, −0.025]	−0.034 [−0.038, −0.029]
W1: EndorseRank (activity) − in-approve degree	+0.018 [0.012, 0.023]	+0.027 [0.021, 0.034]
W1: AWP − transfer in-degree, new senders	+0.020 [0.016, 0.023]	+0.022 [0.017, 0.027]
Contract traders, liquidation-free close rate (registered pairs and rules)		
P, W1, 95% CI	+0.063 [−0.026, 0.148]	+0.084 [−0.013, 0.176]
P, W1, 97.5% CI	+0.063 [−0.040, 0.158]	+0.084 [−0.026, 0.188]
P stratified by closes, W1, 95% CI	+0.113 [−0.021, 0.238]	+0.101 [−0.024, 0.231]
P, W0	+0.252 [0.093, 0.413]	+0.182 [0.052, 0.314]
Q, W1, 97.5% CI	+0.051 [−0.104, 0.232]	−0.041 [−0.148, 0.064]
Q, W0	+0.062 [−0.078, 0.219]	−0.014 [−0.147, 0.130]
EndorseRank − AWP, W1	+0.066 [−0.069, 0.227]	−0.032 [−0.133, 0.066]
EndorseRank − AWP, W0	+0.062 [−0.079, 0.217]	−0.017 [−0.156, 0.128]
C-PR(λ=1) − C-PR(λ=0), W1	+0.171 [−0.083, 0.438]	−0.049 [−0.169, 0.060]
C-PR(λ=1) − C-PR(λ=0), W0	+0.205 [−0.041, 0.437]	−0.041 [−0.187, 0.105]
R1 (count-level lead)	does not hold	does not hold
R2 (PageRank-level lead)	does not hold	does not hold
R3 (specific to liquidation avoidance)	applies	applies

The registered analysis follows the analysis plan and the W1 registration as committed; its values and verdicts are registered results only where a rule of the registration decides them (P, Q, R1–R3); the other trader rows are registered pairs that decide no rule. The revised analysis values amounts in US dollars over five tokens; it was decided after every registered result was known, and its verdicts apply the registered rules to scores defined afterwards. Degree contrasts are not decided by any rule in either analysis. $\Delta\tau$ with 95% interval unless marked 97.5% (spenders 400, traders 2,000 paired resamples, seed 42). Sources: same-window, `holdout-w0`, `registered-w1` and `window-w1` summaries and `supplementary.json` of each analysis in `phd_works`; `analysis/closed-form/closed_form_check.json`; `analysis/neutral-label/results.json`.

its owner's total approval weight. That normalization lowers a spender whose owners approve many other contracts; whether it, or the recency and value weights, cost EndorseRank its agreement with later approvals was not isolated. The variant with activity restarts, which sends restart mass to the owners that approve most, beat the count in both windows. In these comparisons, all made after the labels were known, changing the restart rule moved EndorseRank's agreement with later approval pairs more than the walk moved it away from the count (Table 3); that the restart rule matters more than the walk is a hypothesis. We did not test whether the same holds for AWP, which was run only with its published restarts, and the variant's lead needs a registered test on a window not yet extracted. Such a test should report spenders of depth one, where the walk reduces to a weighted count, separately from deeper spenders, where propagation can matter.

What the trader comparison shows

The trader outcomes are recorded by the exchange and not by either graph, and they were the only outcomes in these data that neither graph is built from, although earlier GMX payouts can appear in the transfer graph. On the primary outcome the comparison did not separate the two edge types. In W1 the approval count ranked later

liquidation avoidance at $\tau = 0.394$ $[0.265, 0.518]$ and the transfer count at 0.310 $[0.184, 0.425]$, and the difference between them, P, had a 97.5% interval that reached below zero. In W0 P was positive with a 97.5% interval above zero, but W0 was registered as the secondary window, its label window lies inside W1's score window, and the rule needed the primary window. The PageRank pair pointed toward the transfer side in both windows of the revised analysis and toward the allowance side in both windows of the registered analysis, without being resolved in any of the four (Table 6). On the outcomes of trading success the transfer side led wherever a difference between an allowance-side and a transfer-side score was resolved, and no score ranked the profitable share with an interval above zero in either window.

What remains is descriptive: no recipient contract was liquidated in either window. That pattern rests on a few dozen contracts, may reflect what those contracts are rather than who approved them, and was not compared with what a venue already knows before the freeze, such as a contract's own liquidation record, its number of closes or its leverage. It does not show value beyond past behavior, and nothing here suggests that approvals measure creditworthiness, default risk or trading skill. Whether an approval count could help a margin or lending venue screen contract counterparties remains open; it would need a registered test on a larger sample, with past behavior controlled.

Implications for evaluating on-chain reputation

The study suggests four practices for evaluating on-chain reputation scores. First, report the raw counts beside every walk: here the transfer walk beat its count and the allowance walk did not, and a walk checked only against statistics of its own edges [16] could not have shown either. Second, freeze scores before any label is counted. Third, compare edge types on outcomes that neither was built from, since each leads on the labels of its own construct by design [20], and check how separate those outcomes are. Fourth, state for every comparison when it was fixed relative to the data, and keep a deviation such as the revision of the amounts visible beside the registered results [41]. Here the revision changed no verdict, but it reversed the sign of the unresolved PageRank pairs on the trader outcome.

Practical implications

An owner deciding whether to approve a contract could be shown how many distinct owners already hold a positive allowance to it, a public number that wallet software can compute; out of window, EndorseRank with uniform restarts ranked later new approval pairs below that number. Such a count tells an owner how widely a contract is already authorized, not what the contract will do with an allowance: a phishing contract gains approvals as an honest router does, and a new contract scores low whatever its code. A low count is a reason to look closer, and a high count is no reason to sign. A raw count of distinct owners is also cheap to inflate, since approvals from N fresh addresses raise it by N for N transaction fees with no tokens locked, and a uniformly restarted PageRank is open to farms of addresses that approve one another [38, 39]; the companion manuscript quantifies that exposure. For ranking which contracts will gain new transfer senders, AWP ranked above the transfer count in both windows (Table 3), in comparisons made after the labels were known. None of these scores is a security audit or a credit score, and none replaces collateral [52].

Limitations

The evidence comes from one chain, one 33-month score window and two three-month label windows, so other rollups and Layer-1 networks would need new data [2]. The

graphs keep five tokens, so contracts that owners approve only on other tokens are outside the cohort, and native ether enters only as WETH. Both graphs are ego networks of one cohort of contracts, so owners' out-strengths, senders' activeness and the paths a walk can take are those of that network and not of the chain, and the finding about propagation does not carry over to graphs with longer endorsement paths. The valuation of amounts was revised after the registered results were known, so the tokens, the price source, the slopes and the cap were all chosen by people who had seen those results; the revision was committed before any revised score existed, its constants come from events up to the first freeze, and the registered analysis reached the same verdicts, which limits that risk without removing it. AWP's parameters are our choices, as its paper gives none. The spender labels are sparse counts of arrivals, which favor any score that tracks the current number of counterparties. Every interval treats contracts as independent and the graphs as fixed, although spenders share owners, so the intervals are probably too narrow; the spender analyses use only 400 resamples, and only the decision pair is adjusted for multiplicity. Account type was read at blocks after both label windows, and a contract that destroyed itself earlier counts as an externally owned account and drops out. An `Approval` event can record the remainder after a spend as well as a grant, and allowances inside managers such as Permit2 are unseen. The trader cohorts are small, the trader labels count partial closes, and market conditions, which drive liquidations, were not measured. No default labels from unsecured lending were available, and no causal claim is made.

Conclusions

Allowance edges and transfer edges carry different information about the same smart contracts. On Arbitrum One, scores built on the latest allowances ranked which contracts would gain new approvals, and scores built on transfers which would gain new senders. In comparisons with the raw degrees, all made once the labels were known, AWP placed the contracts that went on to gain new senders above the transfer in-degree in both windows, while EndorseRank with uniform restarts placed those that went on to gain approval pairs below the count of approving owners in both; for most spenders it is a reweighted approval count, as its closed form shows. A variant with activity restarts beat that count and awaits a registered test. On an outcome that neither graph contains, the later liquidation avoidance of contract traders, the comparison registered for the second window did not separate allowance from transfer at the count level or at the PageRank level, in the revised analysis as in the registered one. The next step is a registered test of the activity-restart variant on a window not yet extracted, with depth-one and deeper spenders reported separately.

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References

1. Schär F. Decentralized finance: On blockchain- and smart contract-based financial markets. Federal Reserve Bank of St Louis Review. 2021;103(2):153-74.

doi: 10.20955/r.103.153-74.

2. Werner S, Perez D, Gudgeon L, Klages-Mundt A, Harz D, Knottenbelt W. SoK: Decentralized finance (DeFi). In: Proceedings of the 4th ACM Conference on Advances in Financial Technologies. ACM; 2022. p. 30-46. doi: 10.1145/3558535.3559780.

3. Vogelsteller F, Buterin V. ERC-20: Token standard. Ethereum Improvement Proposals; 2015. EIP-20. Available from: https://eips.ethereum.org/EIPS/eip-20.

4. Lundfall M. ERC-2612: Permit extension for EIP-20 signed approvals. Ethereum Improvement Proposals; 2020. EIP-2612. Available from: https://eips.ethereum.org/EIPS/eip-2612.

5. Wang D, Feng H, Wu S, Zhou Y, Wu L, Yuan X. Penny wise and pound foolish: Quantifying the risk of unlimited approval of ERC20 tokens on Ethereum. In: Proceedings of the 25th International Symposium on Research in Attacks, Intrusions and Defenses. ACM; 2022. p. 99-114. doi: 10.1145/3545948.3545963.

6. He B, Chen Y, Chen Z, Hu X, Hu Y, Wu L, et al. TxPhishScope: Towards detecting and understanding transaction-based phishing on Ethereum. In: Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security. ACM; 2023. p. 120-34. doi: 10.1145/3576915.3623210.

7. He B, Hu Y, Chen Z, Chen Y, Yu T, Chang R, et al. Unmasking the shadow economy: A deep dive into drainer-as-a-service phishing on Ethereum. In: Proceedings of the 2025 ACM Internet Measurement Conference. ACM; 2025. p. 430-44. doi: 10.1145/3730567.3764476.

8. Adamczyk B. Dissection of an approval-based trusted-token fraud scheme used across multiple blockchain systems. IEEE Access. 2025;13:34467-82. doi: 10.1109/ACCESS.2025.3543651.

9. Lin Z, Chen J, Wu J, Zhang W, Wang Y, Zheng Z. CRPWarner: Warning the risk of contract-related rug pull in DeFi smart contracts. IEEE Transactions on Software Engineering. 2024;50(6):1534-47. doi: 10.1109/TSE.2024.3392451.

10. Human Passport. Human Passport developer platform; n.d. Available from: https://docs.passport.human.tech/ (accessed 9 October 2026).

11. Page L, Brin S, Motwani R, Winograd T. The PageRank citation ranking: Bringing order to the web. Stanford InfoLab; 1999. SIDL-WP-1999-0120. Available from: http://ilpubs.stanford.edu:8090/422/.

12. Langville AN, Meyer CD. Google's PageRank and beyond: The science of search engine rankings. Princeton University Press; 2006. doi: 10.1515/9781400830329.

13. Kamvar SD, Schlosser MT, Garcia-Molina H. The EigenTrust algorithm for reputation management in P2P networks. In: Proceedings of the 12th International Conference on World Wide Web. ACM; 2003. p. 640-51. doi: 10.1145/775152.775242.

14. Gyöngyi Z, Garcia-Molina H, Pedersen J. Combating web spam with TrustRank. In: Proceedings of the Thirtieth International Conference on Very Large Data Bases. Morgan Kaufmann; 2004. p. 576-87. doi: 10.1016/B978-012088469-8.50052-8.

15. Do HD, Do T. PageRank and HodgeRank on Ethereum transactions: A measure for social credit. International Journal of Software Innovation. 2023;11(1):1-13. doi: 10.4018/IJSI.315737.

16. Do T, Do D, Nguyen L. On-chain reputation ranking by Adaptive Weighted PageRank. In: 2023 RIVF International Conference on Computing and Communication Technologies (RIVF). IEEE; 2023. p. 318-23. doi: 10.1109/RIVF60135.2023.10471809.

17. Cong LW, Li X, Tang K, Yang Y. Crypto wash trading. Management Science. 2023;69(11):6427-54. doi: 10.1287/mnsc.2021.02709.

18. Luo J, Kang H, Zheng S, Liu X. Toward resilient airdrop mechanisms: Empirical measurement of hunter profits and airdrop game theory modeling. In: 2025 IEEE International Conference on Blockchain and Cryptocurrency (ICBC). IEEE; 2025. p. 1-5. doi: 10.1109/ICBC64466.2025.11114597.

19. Victor F. Address clustering heuristics for Ethereum. In: Bonneau J, Heninger N, editors. Financial cryptography and data security: FC 2020. vol. 12059 of Lecture Notes in Computer Science. Springer; 2020. p. 617-33. doi: 10.1007/978-3-030-51280-4_33.

20. Cronbach LJ, Meehl PE. Construct validity in psychological tests. Psychological Bulletin. 1955;52(4):281-302. doi: 10.1037/h0040957.

21. Messick S. Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. American Psychologist. 1995;50(9):741-9. doi: 10.1037/0003-066X.50.9.741.

22. Barabási AL, Albert R. Emergence of scaling in random networks. Science. 1999;286(5439):509-12. doi: 10.1126/science.286.5439.509.

23. Newman MEJ. Clustering and preferential attachment in growing networks. Physical Review E. 2001;64(2):025102. doi: 10.1103/PhysRevE.64.025102.

24. Liben-Nowell D, Kleinberg J. The link-prediction problem for social networks. Journal of the American Society for Information Science and Technology. 2007;58(7):1019-31. doi: 10.1002/asi.20591.

25. Lin D, Wu J, Xuan Q, Tse CK. Ethereum transaction tracking: Inferring evolution of transaction networks via link prediction. Physica A: Statistical Mechanics and its Applications. 2022;600:127504. doi: 10.1016/j.physa.2022.127504.

26. Fortunato S, Boguñá M, Flammini A, Menczer F. Approximating PageRank from in-degree. In: Aiello W, Broder A, Janssen J, Milios E, editors. Algorithms and models for the web-graph: WAW 2006. vol. 4936 of Lecture Notes in Computer Science. Springer; 2008. p. 59-71. doi: 10.1007/978-3-540-78808-9_6.

27. Halu A, Mondragón RJ, Panzarasa P, Bianconi G. Multiplex PageRank. PLoS ONE. 2013;8(10):e78293. doi: 10.1371/journal.pone.0078293.

28. De Domenico M, Solé-Ribalta A, Omodei E, Gómez S, Arenas A. Ranking in interconnected multilayer networks reveals versatile nodes. Nature Communications. 2015;6:6868. doi: 10.1038/ncomms7868.

29. Lin D, Wu J, Fu Q, Zheng Z, Chen T. RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. IEEE Transactions on Dependable and Secure Computing. 2025;22(3):2132-45. doi: 10.1109/TDSC.2024.3475825.

30. Wu J, Yuan Q, Lin D, You W, Chen W, Chen C, et al. Who are the phishers? Phishing scam detection on Ethereum via network embedding. IEEE Transactions on Systems, Man, and Cybernetics: Systems. 2022;52(2):1156-66. doi: 10.1109/TSMC.2020.3016821.

31. Bellini E, Iraqi Y, Damiani E. Blockchain-based distributed trust and reputation management systems: A survey. IEEE Access. 2020;8:21127-51. doi: 10.1109/ACCESS.2020.2969820.

32. Almasoud AS, Hussain FK, Hussain OK. Smart contracts for blockchain-based reputation systems: A systematic literature review. Journal of Network and Computer Applications. 2020;170:102814. doi: 10.1016/j.jnca.2020.102814.

33. Hassija V, Bansal G, Chamola V, Kumar N, Guizani M. Secure lending: Blockchain and prospect theory-based decentralized credit scoring model. IEEE Transactions on Network Science and Engineering. 2020;7(4):2566-75. doi: 10.1109/TNSE.2020.2982488.

34. Kotb MO. Credit scoring using machine learning algorithms and blockchain technology. In: 2023 Intelligent Methods, Systems, and Applications (IMSA). IEEE; 2023. p. 381-6. doi: 10.1109/IMSA58542.2023.10217411.

35. Packin NG, Lev-Aretz Y. Decentralized credit scoring: Black box 3.0. American Business Law Journal. 2024;61(2):91-111. doi: 10.1111/ablj.12240.

36. Mukhia K, Luwang SR, Nurujjaman M, Chakraborty T, Saha S, Hens C. Statistical patterns in the Ethereum blockchain: Analysis of EOAs and smart contracts in ERC20 token network. PLOS One. 2026;21(10):e0358972. doi: 10.1371/journal.pone.0358972.

37. Kleinberg JM. Authoritative sources in a hyperlinked environment. Journal of the ACM. 1999;46(5):604-32. doi: 10.1145/324133.324140.

38. Gyöngyi Z, Garcia-Molina H. Link spam alliances. In: Proceedings of the 31st International Conference on Very Large Data Bases. VLDB Endowment; 2005. p. 517-28. Available from: https://www.vldb.org/conf/2005/papers/p517-gyongyi.pdf.

39. Cheng A, Friedman E. Sybilproof reputation mechanisms. In: Proceedings of the 2005 ACM SIGCOMM Workshop on Economics of Peer-to-Peer Systems. ACM; 2005. p. 128-32. doi: 10.1145/1080192.1080202.

40. Douceur JR. The Sybil attack. In: Druschel P, Kaashoek F, Rowstron A, editors. Peer-to-peer systems. vol. 2429 of Lecture Notes in Computer Science. Springer; 2002. p. 251-60. doi: 10.1007/3-540-45748-8_24.

41. Nosek BA, Ebersole CR, DeHaven AC, Mellor DT. The preregistration revolution. Proceedings of the National Academy of Sciences. 2018;115(11):2600-6. doi: 10.1073/pnas.1708274114.

42. Kalodner H, Goldfeder S, Chen X, Weinberg SM, Felten EW. Arbitrum: Scalable, private smart contracts. In: Proceedings of the 27th USENIX Security Symposium. USENIX Association; 2018. p. 1353-70. Available from: https://www.usenix.org/conference/usenixsecurity18/presentation/kalodner.

43. GMX. GMX docs; n.d. Available from: https://docs.gmx.io/ (accessed 9 October 2026).

44. Buterin V, Wilson S, Dietrichs A, lightclient. EIP-7702: Set code for EOAs. Ethereum Improvement Proposals; 2024. EIP-7702. Available from: https://eips.ethereum.org/EIPS/eip-7702.

45. DefiLlama. DefiLlama API documentation. Coins endpoints; n.d. Available from: https://api-docs.defillama.com/ (accessed 10 October 2026).

46. Chainlink. Selecting quality data feeds. Chainlink documentation; n.d. Available from: https://docs.chain.link/data-feeds/selecting-data-feeds (accessed 10 October 2026).

47. Perez D, Werner SM, Xu J, Livshits B. Liquidations: DeFi on a knife-edge. In: Borisov N, Diaz C, editors. Financial cryptography and data security: FC 2021, Part II. vol. 12675 of Lecture Notes in Computer Science. Springer; 2021. p. 457-76. doi: 10.1007/978-3-662-64331-0_24.

48. Kendall MG. A new measure of rank correlation. Biometrika. 1938;30(1/2):81-93. doi: 10.2307/2332226.

49. Efron B. Bootstrap methods: Another look at the jackknife. The Annals of Statistics. 1979;7(1):1-26. doi: 10.1214/aos/1176344552.

50. DiCiccio TJ, Efron B. Bootstrap confidence intervals. Statistical Science. 1996;11(3):189-228. doi: 10.1214/ss/1032280214.

51. Leskovec J, Rajaraman A, Ullman JD. Mining of massive datasets. 3rd ed. Cambridge University Press; 2020. doi: 10.1017/9781108684163.

52. Gudgeon L, Perez D, Harz D, Livshits B, Gervais A. The decentralized financial crisis. In: 2020 Crypto Valley Conference on Blockchain Technology (CVCBT). IEEE; 2020. p. 1-15. doi: 10.1109/CVCBT50464.2020.00005.

Supporting information

S1 Text. Event decoding, valuation, label formulas and extraction queries. Topic hashes and decoded fields of the ERC-20 and GMX V2 events, the account-type rule, the token rule and the US-dollar valuation, exact formulas of every spender and trader label, the SQL patterns, and properties of the data that were not quantified.

S1 File. Pre-specification documents. The analysis plan (commit 43d8045), the W1 registration (commit 0d3a057, SHA-256 recorded in the extraction manifest) and the revision of the amounts (commits a897475, 09dfc32 and 2705b28), reproduced verbatim with their commit identifiers and hashes.

S1 Table. Windows, cohorts, graphs and amounts. Descriptive statistics of the same-window evaluation and both holdout windows, from the matched cohort to the trader recipients.

S2 Table. All contrasts of the comparison on trader outcomes. Every registered pair on all six trader labels in both windows, with 95% and 97.5% intervals and the stratified rows, for the revised analysis and for the registered analysis.